

Multiplexed Single Cell Secretion Assays:
A Microscopy and Machine Learning Approach

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Abstract

Microscopy based multiplexed assays are an important tool for comparing stem cell derived beta cells and donor beta cells through their secretion profiles. The limited fluorescent range used by microscopes limits multiplexed assays to the detection of 3 to 4 biomolecules. This is because fluorescent labels emit a range of fluorescent wavelengths and overlapping ranges lead to false detections. The proposed approach to overcome this limitation is to apply a machine learning model to segment different bead sizes which contain different bioreceptors. After biomolecules bind to the bioreceptors, they can be fluorescently labeled. The fluorescence intensity of each biomolecule can then be calculated using the segmented masks of the different bead sizes. Finally, the mean fluorescence intensities can be converted into concentrations for each biomolecule using standard curves.

Introduction

Pancreatic beta cells are pancreatic cells that secrete insulin – an important biomolecule for body processes such as managing blood glucose levels through cellular glucose uptake. Type 1 Diabetes (T1D) is an autoimmune disease where pancreatic cells are destroyed by the immune system, thus resulting in less insulin production. Current treatments, as discussed by Jang et al.¹, include lifelong insulin injections, and more recently, pancreatic beta cell transplants (figure 1). Donor beta cells are in short supply and thus there is a critical need for new beta cells sources. Recent research has focussed on the differentiation and genetic engineering of pluripotent stem cells into stem cell derived beta cells (SC β) to be used as a new source for beta cell transplants.

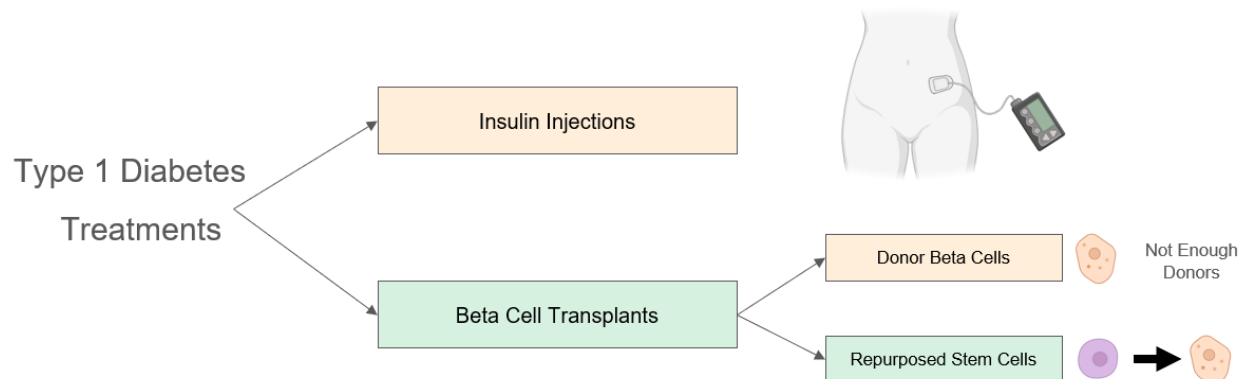


Figure 1: Type 1 Diabetes (T1D) Treatments

Recently, Jang et al.¹ used nanowell technology and fluorescent microscopy to investigate the heterogeneity of glucose stimulated insulin secretion (GSIS) for single pancreatic beta cells. This study provided insight into the characteristics required of stem cell derived beta cells before they can be used as a transplant treatment for Type 1 Diabetes. However, there is still a need for better characterization of donor and SC β cells regarding their insulin, glucagon, and somatostatin hormone secretion levels. This will help in understanding if SC β cells provide the same secretion standards of donor cells (figure 2), or if alternative methods such as secretion-based cell selection if needed.

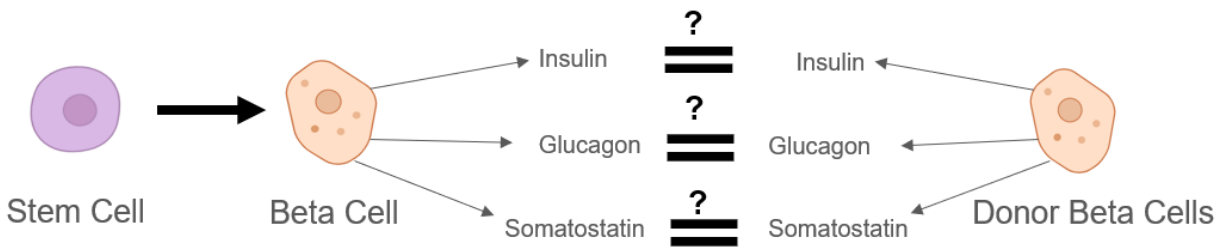


Figure 2: Proposed comparison of donor and SC β cell secretion profiles

Multiplexed single cell secretion assays can be used to profile donor and stem cell derived islet cells through their secretion of insulin, glucagon, and somatostatin. This can be done at the single cell level using nanowells to separate the cells and microscopy to image the fluorescence of detection beads in each nanowell. These detection beads increase in fluorescence intensity when the biomolecule of interest increases (e.g. insulin). However, multiplexed single cell assays that employ microscopy are usually limited to detecting 3-4 different biomolecules². This is due to the limited fluorescence emission range and the fact that fluorophores attached to the beads emit a range of fluorescent wavelengths. Generally, when conducting single cell assays, two fluorescence stains are used to check if the cell is dead or alive. As a result, there is not enough space on the fluorescence emission range for an additional three fluorophores for insulin, glucagon, and somatostatin.

Background

Previous research in increasing the multiplexing capability of microscopy mainly focussed on decreasing the fluorescence emission ranges of fluorescent labels to minimize overlap³. Instead, it is proposed that different bead sizes can be used to detect different biomolecules such as insulin, glucagon, and somatostatin, using the same fluorescent label. Semantic segmentation can be applied to identify and categorize the beads in order to calculate the fluorescent intensity for each bead size.

Semantic segmentation is a well-known challenge in the biomedical field. Recently, researchers have employed deep learning techniques such as U-Net, an encoder-decoder-style neural network, to segment single cells⁴. The U-Net has a proven ability to segment biomedical images with good accuracy and minimal training images using image augmentation techniques⁵. For example, Ryan et al.⁶ applied the U-Net to successfully detect rare sperm in microwell samples from microsurgical testicular sperm extraction. Thus, a U-Net appears to be the best approach towards segmenting different bead sizes in nanowells. In the case of Ryan et al., the classification was conducted on two categories: rare sperm and background. The segmentation discussed in this paper will focus on two different bead sizes and will require the use of three categories, one for each bead size, and one for the background (including cells). In addition, each bead size can be used to detect two different biomolecules through the use of two different fluorophores - resulting in the detection of up to four biomolecules. Table 1 provides an example of an assay setup for sensing insulin, glucagon, and somatostatin. Please note that the fluorophores are still to be decided and so will be named A and B for now.

Bead Size	Fluorophore	Biomolecule
2.8 um	A	Insulin
2.8 um	B	Somatostatin
4.5 um	A	Glucagon
4.5 um	B	N/A

Table 1: Example of a multiplexed assay setup

Segmentation is only the first part of the solution towards increasing the multiplexing capacity of microscopy-based assays through different bead sizes. This is the focus of this paper. The second part requires the application of the segmentation mask to calculate the fluorescent intensities for each bead size in each nanowell. For instance, Harink et al.³ developed a software analysis package that measures the fluorescent intensity through an average of the pixel intensities near microsphere boundaries. The method applied to calculate the fluorescent intensity will be dependent on the segmentation results. Fluorescent intensities of each bead size

will be used along with standard curves to determine the concentration of the biomolecules secreted in each nanowell (figure 3).

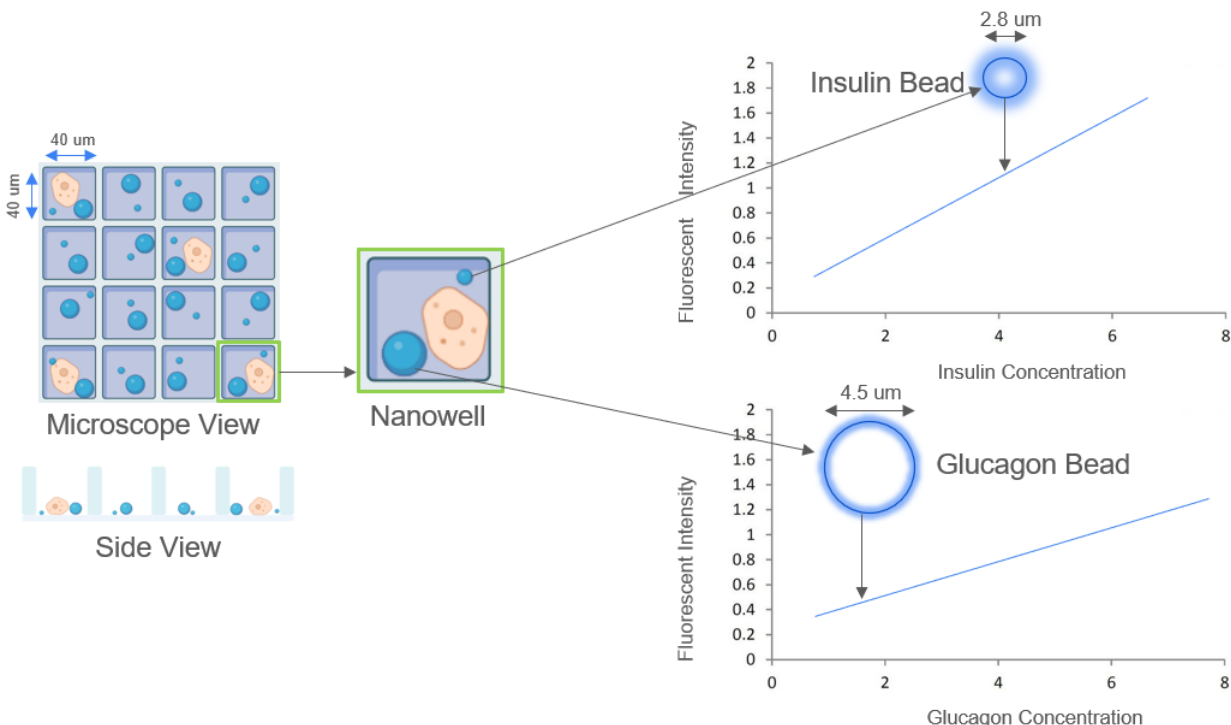


Figure 3: Overview of secreted glucagon and insulin concentration measurements

Methods

Nanowell Partitioning

The Microscopy images obtained are at 20x magnification. These images contain multiple nanowells and thus each nanowell must be cropped into separate images. Morphological close and open operations are used to remove beads and noise to create a grid of white boxes (nanowells). Contours are generated around the white boxes and contour areas between 3,500 and 13,000 pixels are chosen since the nanowell sizes are the same. The center of each nanowell is computed and used to center a box of 160 x 160 pixels for cropping of the entire nanowell. Cropping is completed for brightfield and each fluorescent image using the same centers determined from the brightfield image. Figure 4 provides an example of nanowells being partitioned from a brightfield image and its corresponding fluorescent image. The contour area selection and extent of morphological operations are tuned to maximize the number of nanowells detected while minimizing false detections.

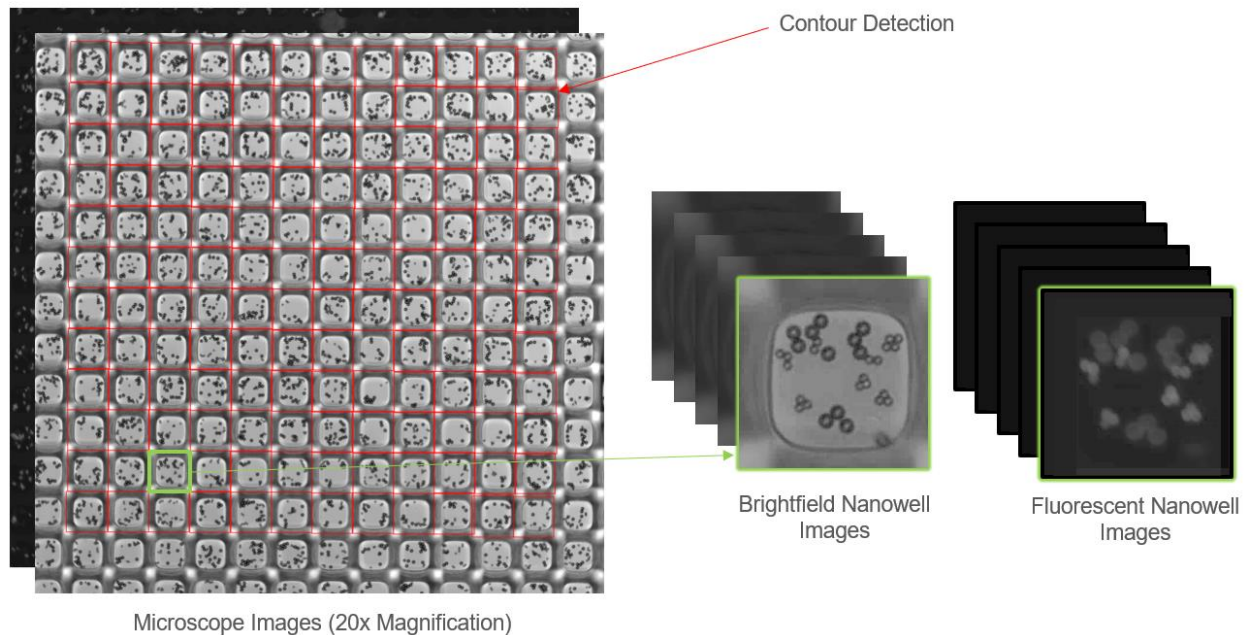


Figure 4: Example of nanowell partitioning

Phase 1: Bead Segmentation

Phase 1 focusses on the setup of the U-Net, training data pre-processing and inspection, and the evaluation of two bead size segmentation using a U-Net. Figure 5 shows the expected result of Phase 1 where the trained U-Net takes a brightfield image of a nanowell and generates a segmentation mask where pixels belonging to 4.5 μm beads are orange, 2.8 μm bead pixels are green, and background pixels are black. This segmentation mask is post-processed to separate the 2.8 μm beads and 4.5 μm beads into separate masks. These masks are binary where the bead pixels are white, and the background pixels are black. This is the main result of Phase 1 and each of the two bead size masks will be used in Phase 2 to measure the fluorescence intensity in the corresponding nanowell fluorescence image at the white pixel locations in the mask. Thus, providing a fluorescence intensity measurement for each bead size that can be used with standard curves to determine the concentration of each biomolecule. Additionally, the masks can be overlaid onto the corresponding brightfield image for a visual inspection of the segmentation.

Phase 1:

Machine Learning Segmentation

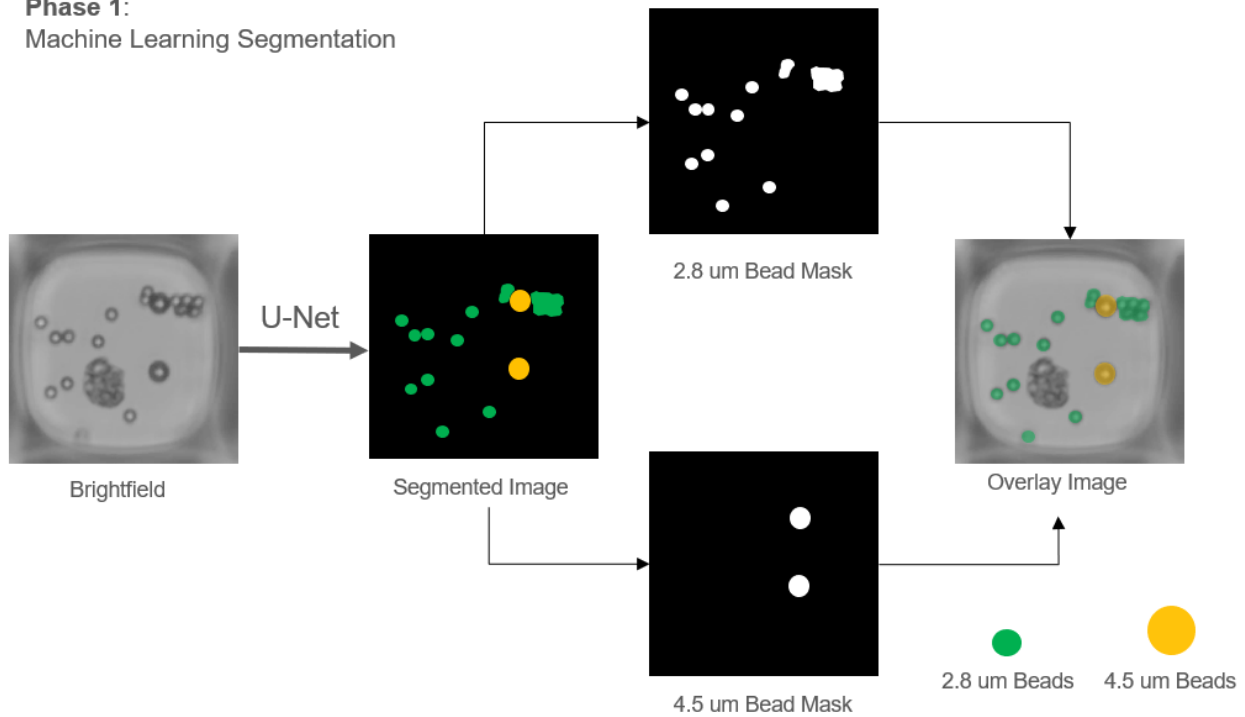


Figure 5: Expected Phase 1 Results - segmentation and mask generation

U-Net Setup

The U-Net is setup in python using Tensorflow's Keras library. The model is currently trained and tested using Google Colab with a T4 GPU. The model takes an input 3-channel 128 x 128 image and outputs a 128 x 128 probability map (figure 1). Input data are normalized between 0 and 1 (inclusive). The encoder applies 2-dimensional convolutions which have a kernel size of 3 x 3, the same padding, and a rectified linear unit activation. The sequential filter sizes applied to the convolutions are 16, 32, 48, 64, and 128. After each 2-dimensional convolution a 2 x 2 max pooling operation is conducted. The decoder applies a series of up-convolutions followed by skip connections where the same shaped output from the encoder is concatenated. After each skip connection, a 2-dimensional convolution is applied with sequential filter sizes of 64, 48, 32, 16. Finally, a 1 x 1 convolution layer with softmax activation is applied to output the probability map. Training is setup for 10 epochs with the Adam optimization and a learning rate of 0.0001. Early stopping, reduce learning rate on plateau, and checkpoint functions in Keras are used to save copies of the model during training and minimize overfitting. The categorical cross-entropy loss function is used to train the model. When maximizing model accuracy is a key objective, model parameters such as activation functions can be changed. Additionally, dropout layers may also be added. The number of epochs, the learning rate, and the loss function may be modified to increase the accuracy and minimize overfitting.

U-Net Training and Validation

In order for the U-Net to generate segmentation predictions, it must be trained to understand its expected output for a given input. Thus, it needs to be provided with brightfield nanowell images as input, and the corresponding two bead sizes (three category) segmentation masks as the expected output (figure 5). The input-output data pairs are generated and inspected before being used to train the model to ensure accurate training. Additionally, image augmentation is used to create more training data pair images for the model to train on.

Ground Truth Generation

Ground truths for the U-Net are nanowell images where each pixel belonging to a specific bead size is classified with a specific colour. The ground truth is another name for the expected output segmentation. For instance, a ground truth image may have the background pixels set to black, the 2.8 μm bead pixels set to green, and the 4.5 μm bead pixels set to orange. These images are paired with their respective brightfield images to train the U-Net. Please note that the ground truths are converted to grayscale where the green pixels are maximum intensity, orange pixels are medium intensity, and the background is black. This is because the designed U-Net outputs a 1 channel grayscale image.

Classical Computer Vision Techniques

Classical Computer Vision (CV) techniques such as thresholding, contour selection, and Boolean operations were initially used to create the ground truths for nanowells with 2.8 μm beads. Although the ground truths are accurate, they are susceptible to false detections in the presence of cells and debris. Additionally, it is unlikely that these techniques would be able to accurately classify pixels of different bead sizes clumped together at 20x magnification.

Fluorescent Techniques

Fluorescent techniques involve the use of fluorophores and fluorescence microscopy to create the ground truths. Each bead size is stained with a different fluorophore. The different fluorophores are chosen to ensure minimal overlap of emission wavelengths – providing accurate ground truths. Two fluorophore colours, green and orange, were used to stain the 2.8 μm and 4.5 μm beads respectively. Fluorescent images are acquired for both fluorophores and are subsequently processed and overlayed onto the brightfield image for quality inspection. The quality inspection reduced the nanowell image data pairs from 2677 to 1858. The main problem observed in the discarded images was background fluorescence that was similar to the bead fluorescence intensity. The overall process is shown in figure 6.

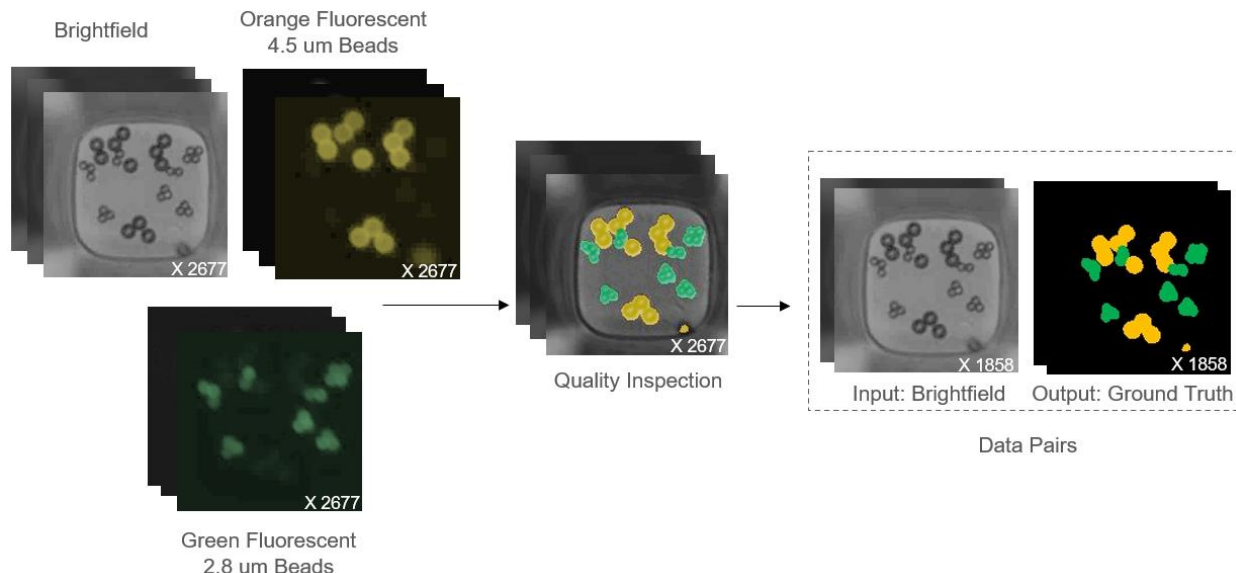


Figure 6: Training data pair generation and quality inspection

Image Augmentation

Generally, 5,000 – 10,000 data pair images are enough to train the U-Net to provide visually accurate segmentation predictions. This is because the backgrounds are very similar since nanowells have the same dimensions. Additionally, since cells are part of the background category, the U-Net mainly has to learn to recognize the beads which are standard and uniform in appearance – thus requiring about 5,000 - 10,000 training image pairs. Please note that the U-Net is not trained on images with cells present as acquiring these images requires experimental protocols and has lead times.

There are 1,858 training image pairs after quality inspection which is below the expected requirement of 5,000 – 10,000 images. Image augmentation is used to generate more training images. This process is more efficient when compared to the lengthy experimentation for acquiring nanowell images. The training image pairs are rotated by 90 degrees, 180 degrees, and 270 degrees. The training image pairs are also flipped horizontally. This is done on 1,700 image data pairs to create 8,500 training image data pairs. The remaining 158 image pairs are used for testing purposes since these are non-augmented nanowell images that are different from the training image pairs. The overall augmentation and distribution of image pairs to training and testing is shown in figure 7. The testing dataset was limited to 158 image pairs to ensure that the training set had sufficient training data. In future training and evaluation of the model, the testing data size will be increased to at least 500 image pairs.

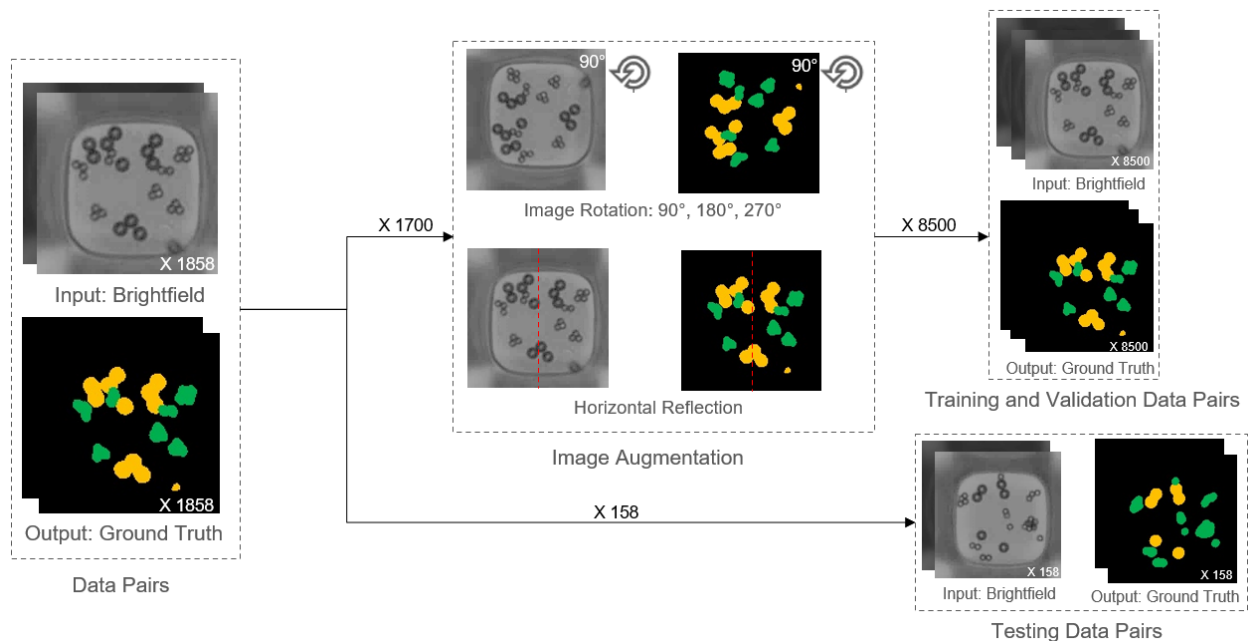


Figure 7: Image augmentation and distribution of training and testing datasets

Results

U-Net Training Results

A total of 8,500 training data pairs are inputted into the U-Net for training and validation. A categorical cross-entropy loss function is used during training. The accuracy and loss are calculated and plotted against the training epoch number. Figure 8 shows that the training loss approaches 0.03 and the training accuracy approaches 0.987. However, this is not as important as the overall curves showing an increase in accuracy and decrease in loss over the epochs – thus showing that the U-Net is learning. Please note that the ground truths inputted into the model were grayscale images where green is max intensity and orange is medium intensity. The ground truths shown in figure 8 are only for visualization purposes.

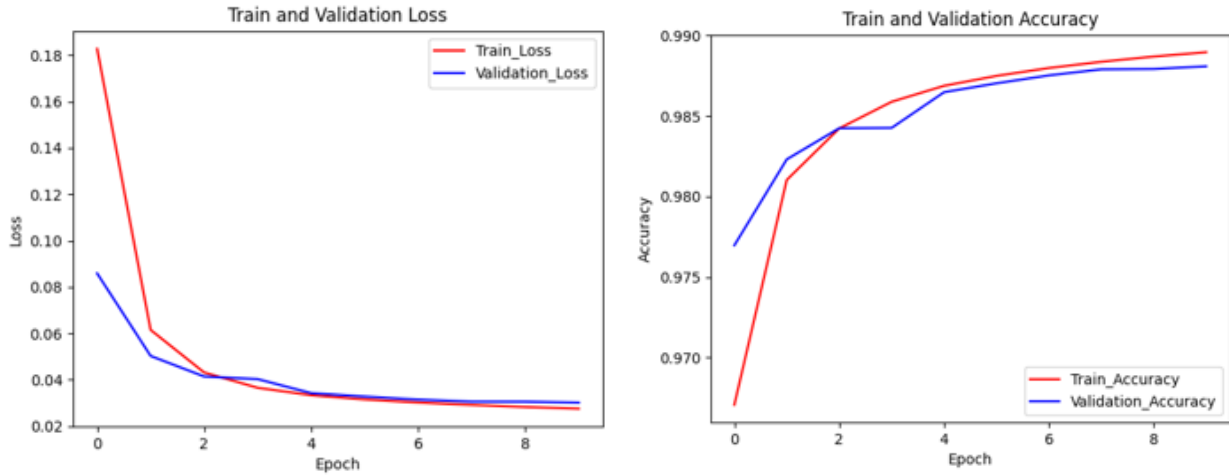
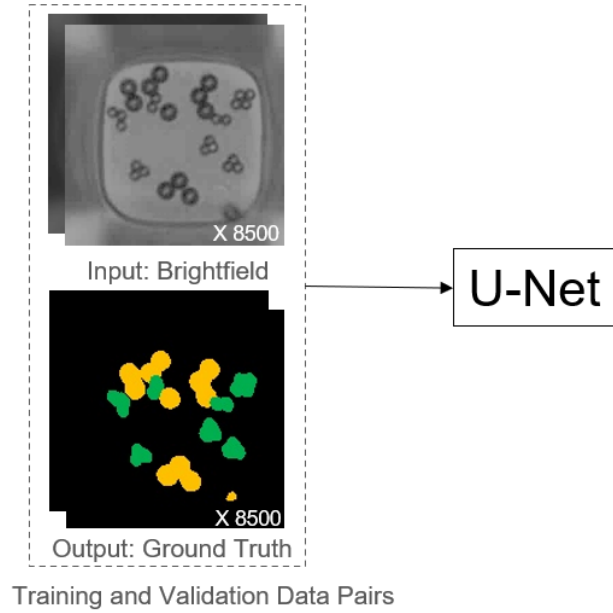


Figure 8: Training and validation of the U-Net Model – Training Loss Graph (left) and Training Accuracy Graph (right)

Test 1: Two Bead Sizes

In the first test, the 158 testing data pairs with the two bead sizes (2.8 μm and 4.5 μm) in each nanowell are used to test the model. The brightfield nanowell images are inputted into the U-Net, which outputs the bead size segmentation predictions (figure 9). The predictions are then compared with the expected output image (ground truth) to determine the accuracy of the segmentation. Please note that the U-Net outputs a prediction that is a grayscale image where 2.8

um beads are high intensity, and 4.5 um beads are medium intensity. The green and orange colors shown in figure 9 are used for visualization purposes.

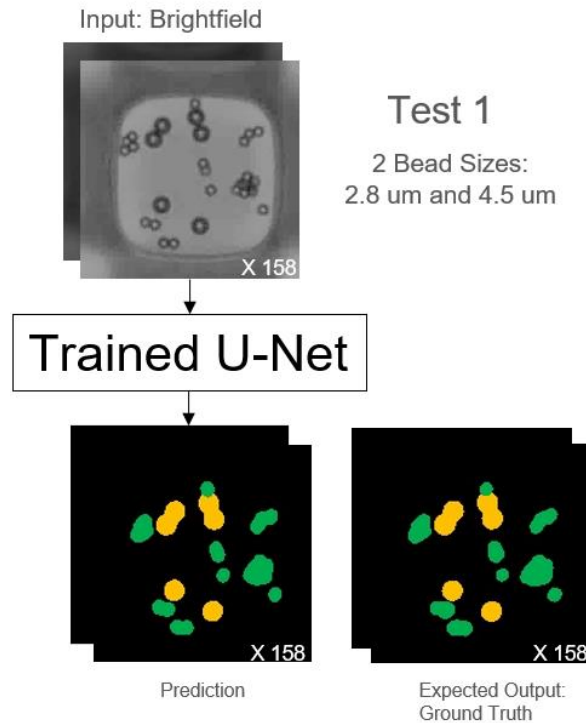


Figure 9: Test 1 – Two bead size segmentation testing

Dice scores are used to evaluate the effectiveness of the segmentation by comparing how well the predicted segmentation matches with the expected segmentation. Dice scores are a standard evaluation metric for segmentation accuracy. A dice score is calculated for each bead segmentation category – 2.8 um beads and 4.5 um beads. Scores near 0 are poor whereas scores near 1 are excellent. Figure 10 provides a graphical representation of the calculation of Dice scores.

$$\text{Dice Score} = \frac{2 \times \text{Intersection Area}}{\text{Predicted Area} + \text{Expected Area}}$$

(Bad) 0 < Dice Score < 1 (Excellent)

Figure 10: Graphical Representation of Dice Score Calculation

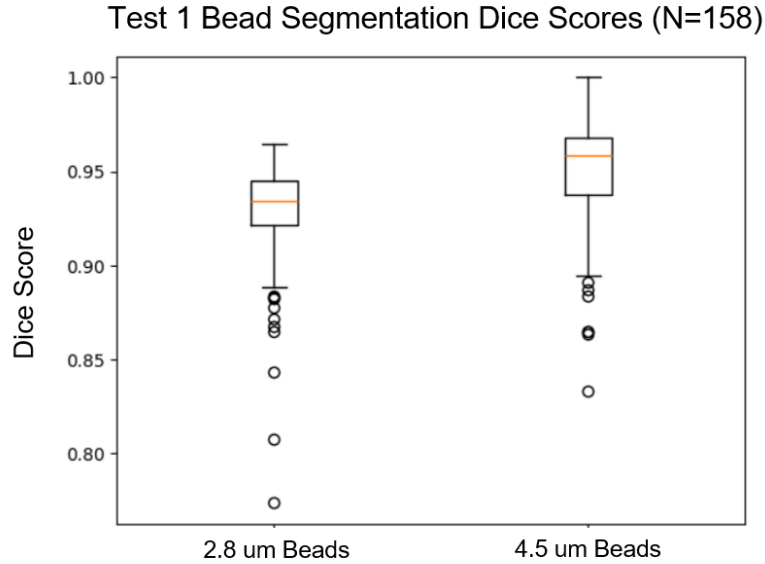


Figure 11: Test 1 Dice Score Results for 2.8 um and 4.5 um beads

The dice scores were calculated for each of the 158 nanowell testing images. The mean dice score for the 2.8 um bead segmentation is about 0.93 and the mean dice score for the 4.5 um bead segmentation is about 0.95. The standard deviations of the scores for both bead sizes is about 0.03 and the minimum score seen is 0.77 for the 2.8 um beads. This shows that most dice scores are above 0.9 – thus showing the model has good accuracy, especially since the standard deviations are small and there are only a few outliers that are near 0.8.

Test 2: One Bead Size with Cell

The previous test used testing data with two bead sizes and no cell present. Since the model is only trained on data without cells present, the model should be tested on images with a cell present to determine if it is able to generalize the cell as background.

Previous experimental images are used to test the model since acquiring new images with cells present incurs a lead time. 18 testing image pairs were found but only had one bead size present (2.8 um bead) along with the cell. The testing pipeline is the same as test 1 (figure 12) but the dice scores are only calculated for the 2.8 um bead segmentation since there were no false positives for the 4.5 um beads in the segmentation predictions.

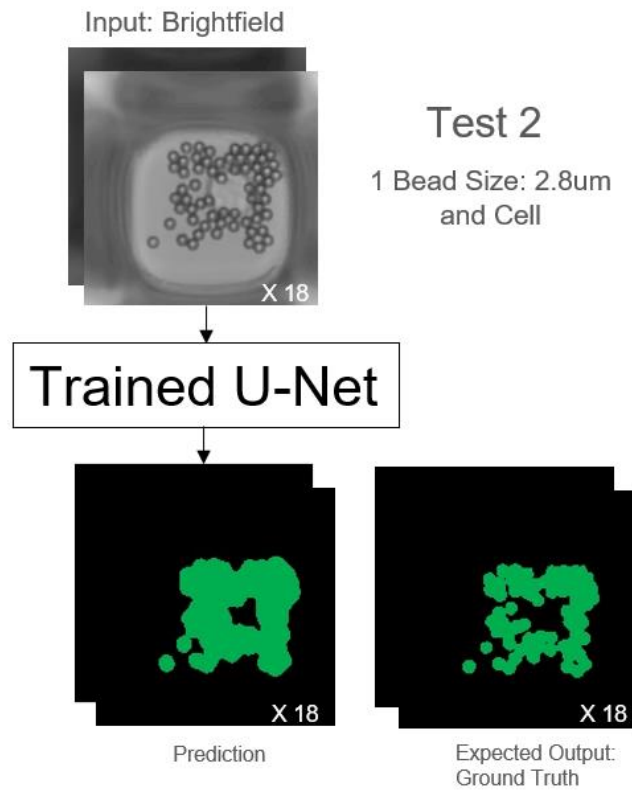


Figure 12: One bead size and cell present segmentation testing

Result (N = 18)	2.8 um Beads
Mean Dice Score	0.66
Dice Score SD	0.25
Min Dice Score	0.18

Table 2: One bead size and cell present Dice Score Results

The calculated Dice score results are shown in table 2. The Dice scores had a mean of 0.66 and a large standard deviation of 0.25. The worst Dice score calculated was 0.18. The results attained are not representative of the overall segmentation accuracy of the model for one bead size and a cell present since only 18 data pairs were used to test the model. Although the testing data size is low, the number of low dice scores (below 0.7) provide sufficient evidence that the model has a low accuracy for segmentation of one bead size with a cell present.

Discussion

The U-Net model was trained on 1,858 image pairs which included an input brightfield image and an output ground truth image (segmentation mask) for single nanowells with two different bead sizes. Additionally, training loss and accuracy graphs indicated that the U-Net learned during training.

Generally, dice scores greater than 0.7 are strong segmentation results⁷. However, larger dice scores near 0.9 are needed since these segmentation masks will be used for quantitative fluorescent intensity measurements. The dice score calculated in the first test (two bead sizes) were mostly above 0.9. A few outliers were as low as 0.77 – thus showing that the trained U-Net model can accurately segment two bead sizes in nanowells. However, since only 158 images were used to evaluate the model in test 1, it is unlikely that this test fully characterizes the accuracy of the model. It is suggested that future tests use at least 500 images for testing to get a stronger characterization of the segmentation accuracy.

The dice scores for test 2 (one bead size and a cell present) were mostly below 0.7 thus showing that the model was unable to accurately segment in this case. This is possibly due to the presence of a cell. It is also possible that the inaccuracy witnessed is due to the higher density of 2.8 μm beads present since the model was trained on images with 4.5 μm beads, which take up additional space. Also, the testing data size was small, only 18 images as these were scavenged from prior experiments. Thus, this test did not completely characterize the model's ability to predict with one bead size and one cell present. However, the results are negative enough to show that the model needs to be trained on a dataset with two bead sizes and a cell present.

The two bead size segmentation is strong enough that it can be used to create masks to measure fluorescent intensity in Phase 2. Thus, the results provide sufficient reason to move towards Phase 2 without the presence of a cell. Cells will be introduced into Phase 2 once the U-Net model is trained on images with cells present, and it can accurately segment the two bead sizes with a cell present.

Currently, the Dice score is the main evaluation metric being applied to assess the accuracy of the segmentation. This was sufficient for the results obtained since the results were either very strong or very weak so that other metrics wouldn't provide much more insight. In the future, additional evaluation metrics such as recall and precision, which characterize the number of false positives and false negatives, can also be used. The overall accuracy of segmentation for this project will be based on the variance in fluorescence intensity measurements across nanowells with the same insulin or glucagon concentration in Phase 2. This is because it is expected that beads with the same biomolecule concentration and an accurate segmentation will have low variance in fluorescence intensity measurements.

Future Work

Image Acquisition and Training with Cells

Future work is split into two main objectives. The first objective is the acquisition of new nanowell images with two bead sizes and a cell present along with the training and testing of the U-Net on the new dataset. This objective will require cell handling training and information pertaining to laboratory procedures will be discussed after training. The image acquisition of these images will be the same process as used for the previous image sets (microscope is pre-programmed). In the case that there is a lead time for the beta cells (donor or stem cell derived), then an alternate cell type can be used if it has similar features when compared to beta cells in brightfield microscopy. Additionally, the data set size needed is about 5,000 – 15,000 image pairs after augmentation – at least 3,000 nanowell images should be acquired (before inspection). Dice scores will be the main metric used for assessment of segmentation. Fluorescent intensity measurements will also be used to characterize segmentation accuracy in Phase 2.

Phase 2: Fluorescent Intensity Measurements

The second objective will focus on applying the U-Net predicted bead masks to the fluorescent bead images to extract the fluorescent intensity for each bead size. The fluorescent intensities can possibly be calculated using techniques such as pixel-based averages for each bead size in the nanowell.

The standard curve investigation is focussed on creating standard curves for two biomolecules using different bead sizes, 2.8 μm and 4.5 μm . Initially each bead size will use a different fluorophore (orange fluorescence and green fluorescence). The biomolecules corresponding to each bead size will be added to each nanowell at different concentrations. Alternatively, one biomolecule concentration can be held constant. Multiple nanowells will have the same concentrations to ensure replicates are created. The different bead sizes will be segmented, and the fluorescence intensities will be measured and plotted against biomolecule concentrations. Each biomolecule will have a standard curve in the case where both biomolecule concentrations are varied.

The variation between replicates and the variability in the fitted curve will be used to assess the strength of segmentation and fluorescence intensity measurements. The standard curve will also be compared to previous standard curves generated using the Nikon Instruments Software. The overall process is shown in figure 13. Once successful, this process will be repeated using the same fluorescence for both bead sizes. This will help in assessing the accuracy of fluorescent intensity measurements and segmentation when both bead sizes have the same fluorescence. This process is shown in figure 14.

Phase 2: Fluorescent Intensity Measurement

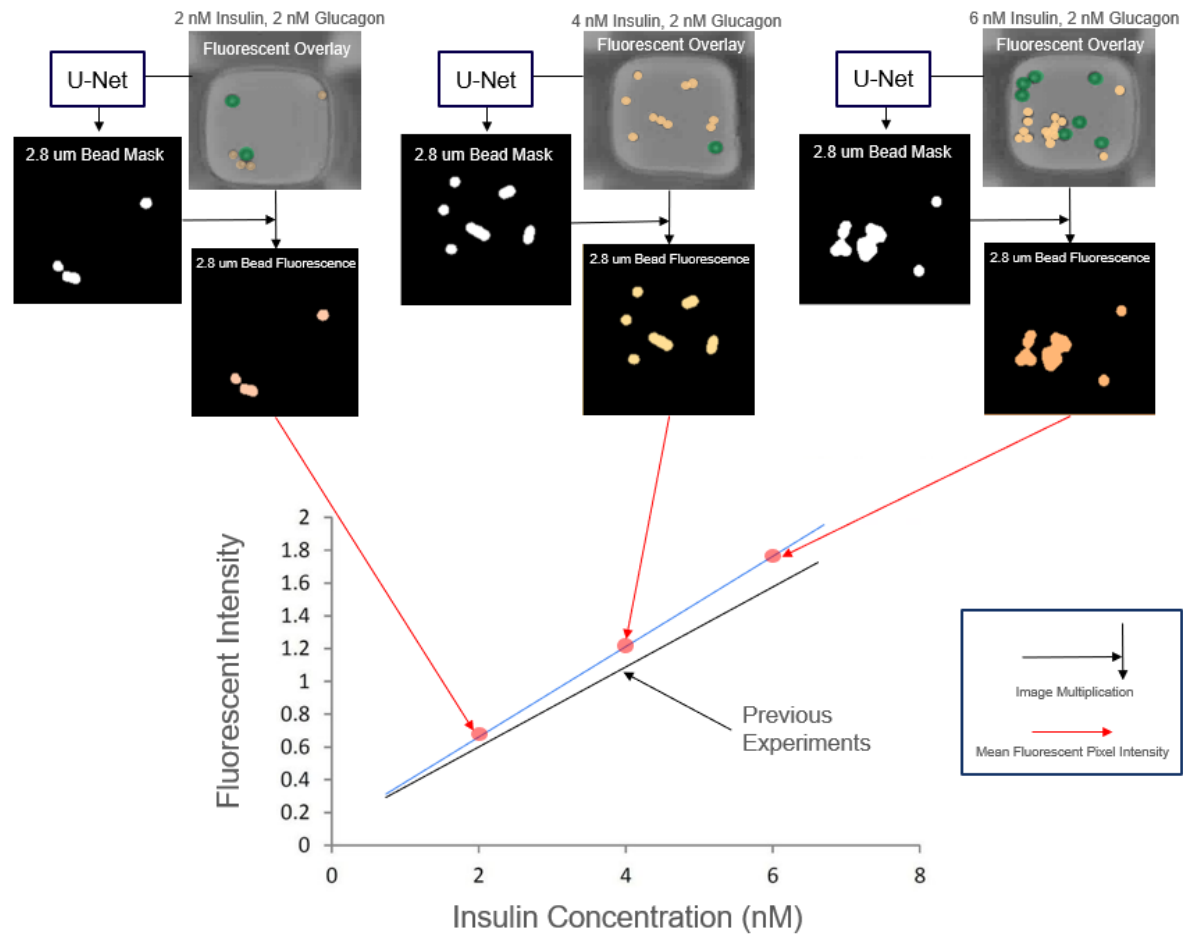


Figure 13: Generation of a calibration curve for insulin through application of U-Net predicted segmentation masks for different bead sizes with different fluorescence

Each fluorescent intensity should only be associated with one concentration on the standard curve. In case of major discrepancies, the nanowell images used to create the curve will be inspected along with segmentation maps. This way it can be determined if the problem is due to the U-Net training data set (Phase 1) or the fluorescent intensity calculations.

The final stage of phase 2 will be a multiplexed single cell secretion assay for pancreatic islet cells. This is still to be planned and is dependant on the results from the standard curve investigation. The main goal of the assay is to measure the concentrations of insulin, glucagon, and somatostatin from either donor or stem cell derived islet cells.

Phase 2: Fluorescent Intensity Measurement

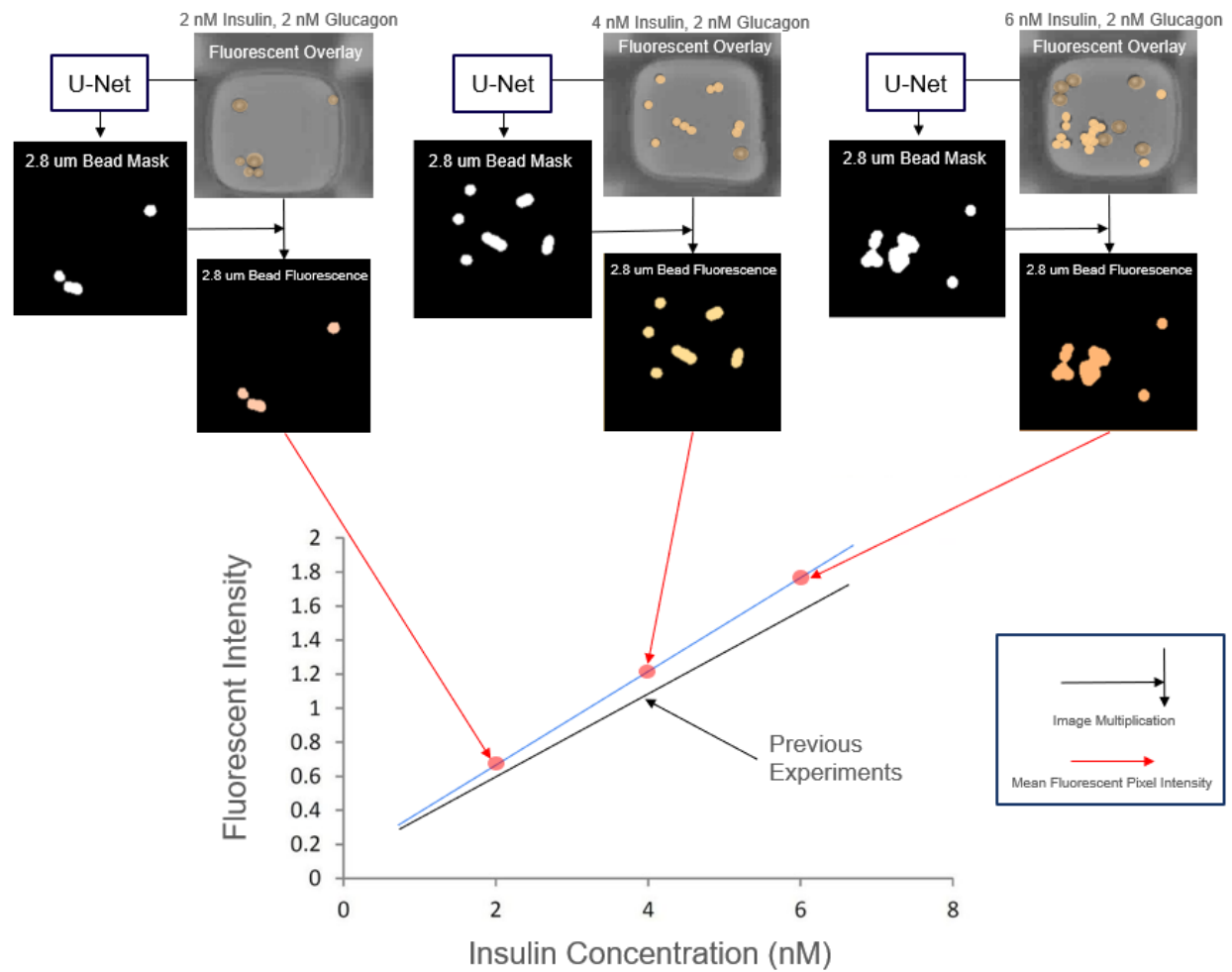


Figure 14: Generation of a calibration curve for insulin through application of U-Net predicted segmentation masks for beads with the same fluorescence

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